Fiscal Year:	FY 2023	**Task Last Updated:**	FY 11/02/2023
PI Name:	Santa Maria, Sergio Ph.D.		
Project Title:	Acquisition of Beneficial Mutations Through Adaptive Evolution Under Simulated Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	650-604-1411
Organization Name:	NASA Ames Research Center		
PI Address 1:	Space Biosciences Research Branch		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	06/01/2021	**End Date:**	12/31/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2024 per F. Hernandez/ARC (Ed., 10/27/23) NOTE: End date changed to 12/31/2023 per F. Hernandez/ARC (Ed., 6/2/23) NOTE: End date changed to 02/28/2023 per F. Hernandez/ARC (Ed., 4/25/22)		
Key Personnel Changes/Previous PI:	Co-Investigator (Co-I) Lauren Liddell is no longer part of the project due to maternity leave starting mid-2022. An undergraduate student intern was recruited in the Fall 2022, Spring 2023, and Fall 2023 to support the Principal Investigator (PI).		
COI Name (Institution):	Sherlock, Gavin Ph.D. (Stanford University)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Adaptive evolution plays a major role in the appearance of new phenotypes in nature and is of key importance in biomedical research, including the onset of carcinogenesis and in the emergence of drug-resistant microorganisms. The harsh environment of space provides the ideal environment for the acquisition of adaptive mutations, particularly in microbial organisms. Experimental evolution studies in the laboratory will allow us to better understand how specific mutations arise under spaceflight conditions. Our primary goal is to investigate how a eukaryotic organism responds to simulated microgravity after multiple generations. S. cerevisiae or budding yeast is a single-cell model eukaryote that has been widely used in biomedical research, primarily due to its well-characterized biology, easy manipulation, short division time, and, most importantly, its similarities with human cell functions. We hypothesize that beneficial mutations acquired through multiple generations under simulated microgravity will render cells with increased growth and/or metabolic rates, and phenotypes different to those observed in 1g controls. Aim1 involves the identification of adaptive beneficial mutations via experimental evolution and next-generation sequencing. Barcoded yeast libraries will be grown using rotating wall vessels (RWVs) for multiple generations. DNA barcodes are unique 20-nucleotide sequences that are used to “tag” or “barcode” individual strains and allow us to follow the appearance of these adaptive mutations and to identify which mutations are actually beneficial to survive under simulated microgravity. The frequency of the resulting yeast lineages and the appearance of adaptive mutations can be then followed by genomic DNA extraction followed by amplification and sequencing. Aim 2 involves the generation of mutational spectra from the surviving yeast at different time points during multigenerational growth. We propose to perform mutation analyses via whole genome sequencing to generate mutation spectra after long-term exposure to simulated microgravity. The resulting acquired mutations will be then compared to those of the non-exposed 1g controls and to previous generation spectra. All RWV growth studies will be performed at Ames Research Center. Barcoded yeast libraries developed by the Sherlock team at Stanford University are already available. DNA amplification and next-generation sequencing will be performed at Stanford. All project data will be archived in the NASA GeneLab repository. Studies like the one proposed here will allow us to identify beneficial mutations responsible for an adaptive response to space-like conditions. The proposed work will address questions on how spaceflight affects microbial growth and population dynamics (Space Biology Microbiology objectives A-C), and how it alters DNA function and intracellular processes (Cell and Molecular Biology objectives B and E). We will try to identify what processes are influenced by the spaceflight environment, and whether long-term exposure has biological effects across multiple generations (Guiding Questions MB-1, MB-2, MB-5, MB-6, CMB- 1, CMB-7, and Decadal Survey recommendations P1, P2, CC9, and AH16, https://).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The significance of exploring the effects of the space environment on microbes (and their adaptation) has important implications. For example, abnormal cell growth and physiology – especially in pathogens – could be detrimental to astronaut health, especially considering NASA’s upcoming long-term missions to outer space. On the other hand, altered microbial growth in space could be advantageous for the production of high-value products, including medicines, vitamins, and food.
Task Progress:	Aim 1: To investigate the acquisition of adaptive beneficial mutations via experimental evolution and high-throughput sequencing. For the past year, we have performed a variety of experiments to determine the best growth conditions for the evolution experiments in standard lab conditions and in rotating wall vessels (RWVs, simulated microgravity). These experiments include growth medium optimization (e.g., sugar concentration), RWV rotation speed, initial inoculate cell density, growth and/or duplication rate, and proper RWV transfer. Importantly, the 21-day RWV evolution experiment was initiated in early October and completed on October 27. Samples were collected at approximately every 8 generations (i.e., every 24 hours) and stored at either -20°C (sorbitol solutions) or -80°C (glycerol stocks). No major issues were encountered during the evolution experiment. The samples will be transferred to the Co-Investigator team at Stanford University, where the frequency of the resulting yeast lineages and the appearance of adaptive mutations (after ~168 generations) will be then followed by genomic DNA extraction, amplification, and sequencing. Aim 2: To explore the mutational spectrum from the surviving yeast at different time points during multigenerational growth. Per the original proposal, we will perform mutation analyses via whole-genome sequencing to generate beneficial mutational spectra after continuous exposure to simulated microgravity in RWVs. The resulting acquired mutations will be then compared to those of the non-exposed 1g control and to previous generations’ spectra.
Bibliography Type:	Description: (Last Updated: 12/08/2025)
Articles in Peer-reviewed Journals	Rahmanian S, Slaba TC, Braby LA, Santa Maria SR, Bhattacharya S, Straume T. "Galactic cosmic ray environment predictions for the NASA BioSentinel mission." Life Sci Space Res. 2023 Aug 1;38:19-28. https://doi.org/10.1016/j.lssr.2023.05.001 ; PMID: 37481304 , Aug-2023
Articles in Peer-reviewed Journals	Liddell LC, Gentry DM, Gilbert R, Marina D, Massaro Tieze S, Padgen MR, Akiyama K, Keenan K, Bhattacharya S, Santa Maria SR. "BioSentinel: Validating sensitivity of yeast biosensors to deep space relevant radiation." Astrobiology. 2023 Jun 1;23(6):648-56. https://doi.org/10.1089/ast.2022.0124 ; PMID: 37052477; PMCID: PMC10254971 , Jun-2023
Articles in Peer-reviewed Journals	Santomartino R, Averesch NJH, Bhuiyan M, Cockell CS, Colangelo J, Gumulya Y, Lehner B, Lopez-Ayala I, McMahon S, Mohanty A, Santa Maria SR, Urbaniak C, Volger R, Yang J, Zea L. "Toward sustainable space exploration: A roadmap for harnessing the power of microorganisms." Nat Commun. 2023 Mar 21;14(1):1391. https://doi.org/10.1038/s41467-023-37070-2 ; PMID: 36944638; PMCID: PMC10030976 , Mar-2023
Books/Book Chapters	Santa Maria SR. "Microbial biology on CubeSats." in "Next Generation CubeSats and SmallSats." Ed. F. Branz, C. Cappelletti, A.J. Ricco, J.W. Hines. Cambridge, MA: Elsevier, 2023. p. 645-54. https://doi.org/10.1016/C2020-0-00508-6 , Aug-2023